An Assessment of the Site Investigation for the North East line, based on a review of Contractors' Claims

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ABSTRACT: Planning a site investigation programme involves a number of factors, including balancing the cost of the investigation against the potential cost of claims if inadequate investigation is carried out. This paper summarises the results of a study into the site investigation carried out for the North East Line. The amount of investigation varied along the alignment, and the amount of investigation is compared with the amount claimed by contractors for unforeseen ground conditions. The results are compared with other published data. It is found that the overall number of boreholes was close to the optimum amount. A more detailed evaluation was then carried out on the relationship between the claims and the ground conditions, the methods of construction and the details of the boreholes. It was found that the majority of claims related to the behaviour and strength of rock, rather than soft ground. Boreholes in rock were often stopped short of the tunnel level, and there was a strong correlation between the proportion of boreholes terminated above the tunnel and the number and value of claims. The results of the study are used to assess how to further improve site investigation for tunnelling in Singapore.

1 INTRODUCTION

It is important to have a good understanding of the ground conditions likely to be encountered before commencing a tunnel project. The ground conditions have a major impact on the choice of tunnelling method and machinery, advance rates and maintenance costs.

It is common practice for the project owner to undertake site investigation, and include the results of that investigation in the tender documents for a tunnel. This allows tenderers to estimate the likely cost of doing the work.

There is a risk that the ground conditions will be significantly different to those assessed from the information available at time of tender. The difference between the assessed and actual ground conditions may drive up the cost of the work, for example if:

- The advance rate is slower than expected
- The cutting tools wear out, and have to be replaced, more often than expected
- There is more water than expected, leading to greater pumping costs and/or instability of the ground.

The most adverse possible change is where the chosen methods and equipment are found to be incapable of allowing safe tunnelling through the revealed ground conditions. This can lead to huge delays while the appropriate equipment is procured and installed.

Because tunnelling is so critically linked to the nature of the ground conditions, it is common, but not universal, for owners to retain the risk of encountering ground conditions more adverse than could reasonably have been anticipated, and not pass this risk on to the contractor.

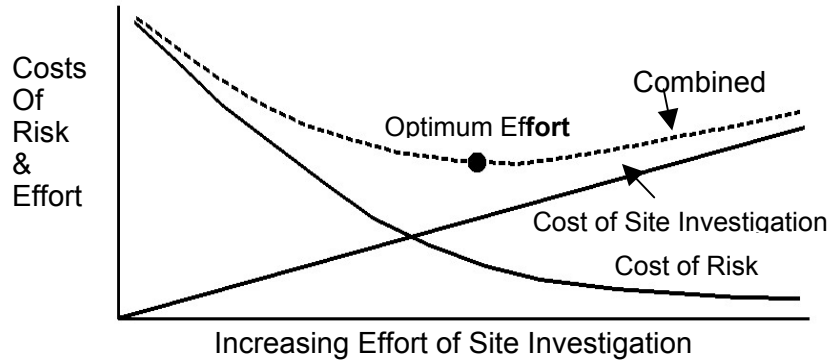


Figure 1. Assumed relationship between cost of Site Investigation and cost of risk.

Conventionally, site investigation in advance of tunnelling involve boreholes, from which samples are obtained for laboratory testing, supplemented where appropriate by insitu testing and geophysical testing. The owner is faced with deciding how much investigation to do. The more investigation that is done, the more knowledge that is obtained about subsurface conditions. However, the amount of knowledge per unit of work is likely to decrease throughout the programme, i.e. there are diminishing returns. The investigation will itself be a cost to the owner, so there must be a point where further investigation is no longer economic. Diagrams such as Figure 1 are commonly postulated to demonstrate this.

In 1984 the U.S. National Committee on Tunnelling Technology published the results of a three year study of subsurface explorations (USNCTT, 1984). Among the many recommendations of this report, there were two that related specifically to the amount of investigation to be allowed for. The committee recommended that the amount spent on investigation should be increased to an average of 3% of predicted construction cost. They pointed out that the average cost, at that time, was 1% of project cost while the average claims payment was 12%. They also recommended that, on average, 1.5 feet of borehole should be drilled for every 1 foot of tunnel.

The recommendations of the US committee give valuable benchmarks against which to assess the appropriate level of investigation for tunnelling. However, the recommendations are related to project experience in the United States, and are therefore based on the US industry cost structure and legal environment. It is therefore useful to assess local experience in order to identify the appropriate levels for investigation.

Civil construction for Singapore's North East line (NEL) started in 1998. The main excavations and tunnelling were completed by December 2000. The NEL contracts included an unforeseen ground conditions clause, and a number of claims were registered under this clause. All of these claims were settled by August 2002.

Based on the NEL data, a study was made of the relationship between the site investigation and the claims for unforeseen conditions, to identify where improvements in the planning for site investigation might usefully be made. This paper summarises the results of that study.

2 PROJECT DETAILS

2.1 General

The NEL is a heavy mass transit rail line, including 16 underground stations, linked by tunnels, and a surface depot. The route is shown in Figure 2. The civil construction work was divided into twelve contracts, let on a design/build basis. Six of the civil contracts included sections of bored tunnelling. There was no contract in which tunnelling formed the preponderance of the work; in all cases the cost of the cut-and-cover construction for stations and tunnels exceeded the cost of bored tunnelling.



Figure 2. Route of the North East Line

The NEL has a route length of approximately 20 kilometers. 8.6 kilometers were constructed using cut-and-cover methods, including all 16 stations. The remaining 11.4 route kilometers involved bored tunnelling, with twin 5.8m diameter tunnels. Of the 22,830m of bored tunnelling, just over 20,000m was constructed using 14 earth pressure balance machines (EPB). Almost all of the remaining tunnelling was carried out using two open-face shields. Details of the shields are given in Reilly and Tham (1999). The tunnels were lined with one pass, reinforced concrete, segmental linings.

2.2 Site investigation and other geotechnical data.

A few boreholes along the route of the NEL were drilled in 1986, during the first feasibility study for the line. The main investigations were carried out in 1995, and consisted primarily of boreholes drilled by rotary methods using washboring. A limited number (128) of cone penetration tests (CPT) were also carried out. The main investigation programme was divided up into three contracts, each let to a different specialist investigation contractor. A limited programme of additional investigation was carried out in 1996 and 1997 to provide supplementary information in areas where the horizontal or vertical alignment had been changed. This information was provided to tenderers, although some information was provided during tender addenda rather than at the start of the tendering period. At two points, at Dhoby Ghaut and Outram Park, the NEL crossed the existing North-South line, which was built in the early 1980's. Site investigation information and the available geological profiles from the construction of the earlier work were made available to tenderers. A search was made for borehole records from adjacent construction sites, and these boreholes were also made available to tenderers. However, many of these latter boreholes had stopped short of the level of the NEL tunnels, and contained limited testing relevant to likely tunnelling conditions.

2.3 Ground conditions

The ground conditions encountered, and the direction of the bored tunnel drives, are summarised in Figure 3. The north eastern half of the route was predominantly in Old Alluvium, although a buried ridge of Bukit Timah granite was encountered in the area of Serangoon Station. The south-western half of the route was mainly in the sedimentary rocks of the Jurong Formation. The mudstone, siltstone, sandstone, conglomerate and limestone beds of the Jurong formation have been subjected to a varying degree of metamorphism. The Old Alluvium, Bukit Timah granite and Jurong Formation have all been subjected to intense tropical weathering to depths of 30m or more. Short sections of the tunnel were driven through the Fort Canning Boulder Bed, which was encountered on the western side of Fort Canning Hill and the eastern side of Mount Emily. Incised into the Old Alluvium and the Jurong Formation were a number of buried valleys, often steeply sided, which were infilled with the soft, recent soils of the Kallang formation. Further information on the soils and rocks is given in Shirlaw et al (2000) and Shirlaw (2002).

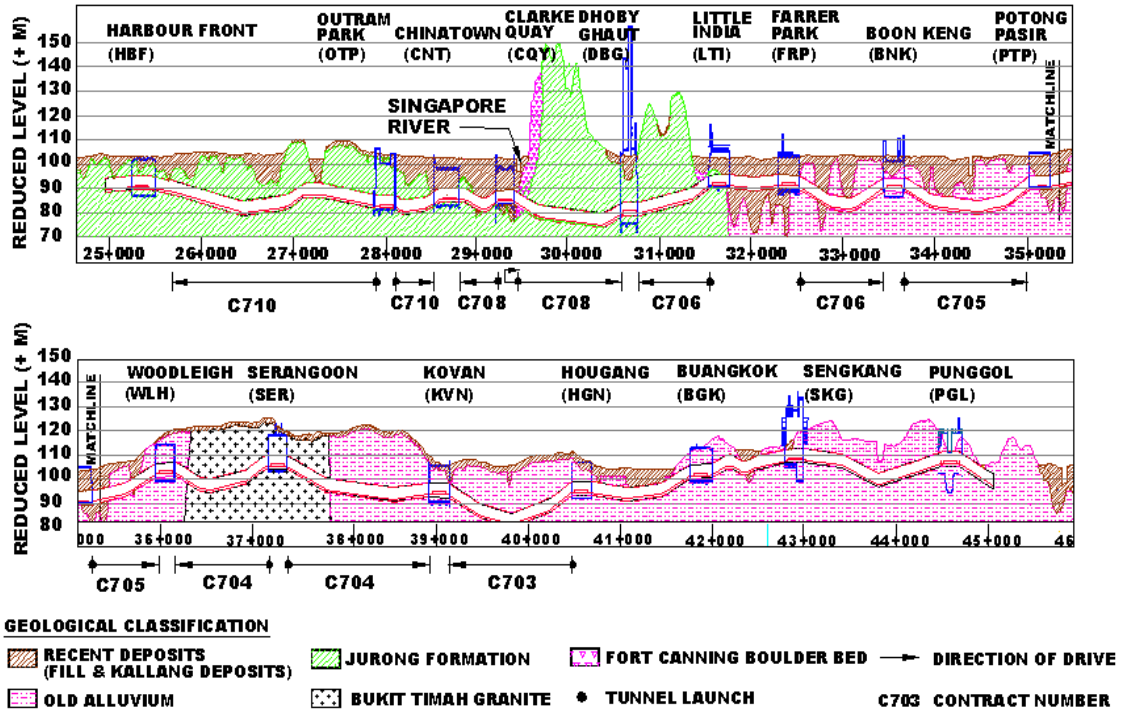


Figure 3. Geological long section along the North East line, showing the bored tunnel sections and direction of the drives.

Tenderers were required to assess the site investigation data in the context of published geological and geotechnical information. For geological conditions the geological map of Singapore, published by the PWD in 1976, is a standard reference. The experience gained during the construction of the earlier phases of Singapore's Mass Transit system was documented in a number of published papers (ie. Hulme et al 1990, Hulme and Burchell 1994). Numerous papers have also been written about basement construction in Singapore, and a more limited number on the construction of sewer tunnels.

3 BASIS OF STUDY

The first step in the study was to define the input data needed for the study. How costs and claims are defined has a major impact on the numbers used.

3.1 Project Cost

The NEL contracts involved stations, as well as tunnels, and included architectural finishes and some elements of the electrical and mechanical equipment needed for railway operation. The contractors' activity costs were used to establish an estimated value for just the civil component of the works, separated between cut-and-cover and bored tunnelling work. It was this civil component that was used to establish percentage values for site investigation and claim values. All costs related to the depot were excluded, as the focus of the study was on site investigation for underground works.

3.2 Site Investigation cost.

The site investigation carried out for the NEL was let as a number of separate contracts, which generally covered several of the main civil contracts. The average cost per meter of borehole, including all testing, mobilisation and overheads, was estimated by dividing the cost of each contract, excluding penetration testing, by the meterage drilled. No allowance was made for the LTA staff involved in managing, supervising and interpreting the work. Only those investigation costs incurred by the Authority specifically for the NEL were included; no allowance was made for boreholes drilled by others or for the earlier phases of MRT construction, and made available to tenderers. The costs were then

apportioned between the bored tunnel and cut-and-cover sections strictly on the basis of chainage; no allowance was made for the benefits obtained from the boreholes drilled for the cut-and-cover structures in terms of understanding the conditions for the adjacent tunnels. During the planning stage some changes had been made to the NEL alignment. As a result, some boreholes were drilled some way from the final alignment. All boreholes more than 50m from the alignment were excluded when assessing the cost and number of the borehole programme. Boreholes for the depot were also excluded. All of the costs relate to investigations carried out by the LTA prior to or during tendering. No account has been taken of the, often significant, investigation programmes carried out by the contractors after award.

3.3 Claims Included.

The study did not include any claims which were registered by the contractor, but which were withdrawn before, or not pursued to the point where, the contractor submitted a breakdown of the additional costs for which he was claiming.

3.4 Claims Value

Many of the claims were settled by commercial agreement, which took into account all of the contractor's claims for additional payment in a single agreement. It was therefore not possible to properly quantify the true commercial value of the claims for unforeseen ground conditions. The study therefore used the contractors' evaluation of claim value as a means of measuring the relative value of the different claims submitted. The true cost of the claims, in terms of the total sum required to settle them, was much smaller than numbers used in the study.

3.5 Artificial obstructions or unforeseen natural ground conditions?

The standard 'unforeseen ground conditions' clause relates to both artificial obstructions and unforeseen natural ground conditions. Although there were a number of claims lodged for artificial obstructions, most of these related to the temporary works installed by the adjoining contractor. These claims were generally withdrawn before the contractor had even assessed the value of them, and are therefore not included in the total claim value presented here. The proportion of the claims value related to artificial obstructions was so small as to be negligible. Effectively, the claims presented here relate to – unforeseen natural ground conditions.

3.6 Length of tunnel constructed.

One way of assessing the overall amount of investigation for tunnels is to divide the total length of the boreholes by the total length of the tunnel. The published data is largely derived from single tunnels. The tunnels for the NEL consisted of parallel running tunnels, typically 10m to 16m apart. Boreholes along the alignment provided information about both tunnels. In order to provide comparison with the historical data, the twin tunnels are treated as one for the purposes of establishing the ratio of the borehole length to the length of tunnel. However, for the assessment of the length of tunnel in a particular formation, for the costs and claims per kilometer of tunnel the figures given are based on the actual length of tunnel driven i.e. the twin tunnels are treated separately.

4 RESULTS OF STUDY

The first step in the study was to assess tabulate all of the claims and civil costs on the NEL, and separate these items between cut-and-cover structures and bored tunnels. The site investigation carried out by the authority was also tabulated and costs assigned, and separated between bored tunnels and cut-and-cover structures. Once this initial overview was completed, the data was further subdivided to assess the various factors related to the claims.

4.1 Results of overview

The 'unforeseen ground condition' claims, as evaluated by the contractors, are summarised in Table 1, and compared with the base cost for the civil works. It can be seen that, relative to civil value, the

Table 1. Amount claimed by contractors for 'unforeseen' condition

	Amount claimed Cost, S$ million	Civil Cost S$, million	Amount claimed % of Civil Cost
Cut-and-cover	22.7	1,136.98	2.0
Bored tunnel	38.4	430.82	8.9
Total	61.1	1,567.80	3.9

claims for the bored tunnelling sections were over four times those on cut-and-cover sections. The ratio of claims made to civil cost for the bored tunnelling was about 9%. This is based on the contractor's evaluation; the actual final cost can only be estimated, but was probably less than 3%. This can be compared against published charts, such as that presented by Westland et al. (1998), and summarised in Figure 4. This chart would suggest that it is reasonable to expect a best estimate, average, 'risk' cost of claims of about 5% of project value, with an optimum site investigation programme. The paid cost of claims for the NEL were certainly below this figure, which suggests that the risk of claims was well managed, with an effective site investigation programme. However, the values presented by Westland et al. are for work in North America, where contractors and owners tend to have been more confrontational and litigious than in Singapore. Even the actual payments for claims on the NEL represent tens of millions of dollars of public money, so it is useful to study the claims and see if lessons can be learned on how investigation planning might be further improved.

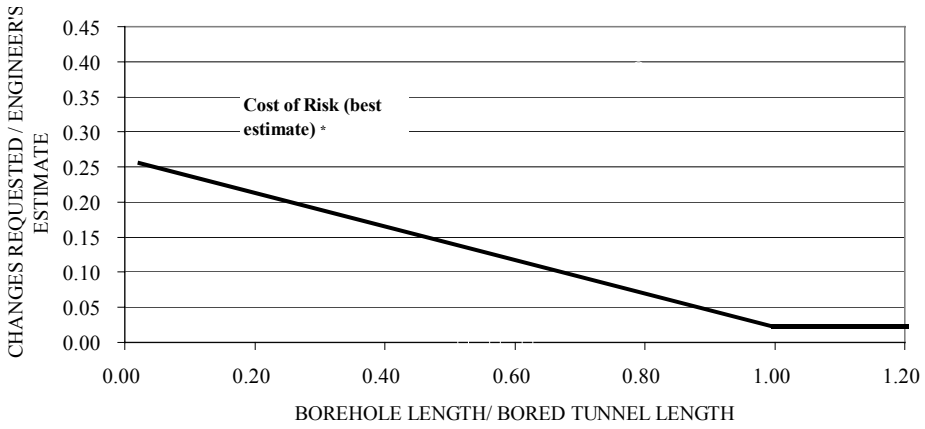


Figure 4. Cost of risk of changes (claims) plotted against relative length of boreholes, after Westland et al (1998)

Table 2 summarises the information on the cost of the site investigation. It can be seen that, the percentage spent, relative to the cost of the civil works, was 0.216%. The percentage spent on the tunnels, at 0.422%, was relatively higher than that spent for the cut-and-cover structures, although still much lower than the 3% recommended by the U.S. National Committee on Tunnelling Technology. In terms of length drilled, Table 3, the ratio of borehole length to route length was 0.6 for the tunnels and 0.89 for the cut and cover structures. This was lower than the 1.5 ratio proposed by the U.S. National Committee on Tunnelling Technology, or the 1.0 ratio suggested by Westland et al. (1998).

Table 2. Ratio between cost of site investigation and civil cost

	Site investigation Cost, S$ million	Civil Cost S$, million	SI Cost % of Civil Cost
Cut-and-cover	1.56	1,136.98	0.137
Bored tunnel	1.82	430.82	0.422
Total	3.38	1,567.80	0.216

Table 3. Ratio between number and length of boreholes and route length, pretender information.

	Boreholes Number	Borehole length, m	Route length m	Average Borehole spacing, m	Borehole length/ route length
Cut-and-cover	262	7,658.5	8,600	32.8	0.89
Bored tunnel	286	6,839.0	11,400	39.9	0.6
Total	548	14,497.5	20,000	36.5	0.72

The ratio of investigation cost to civil construction cost appears very low relative to the ratio of meterage drilled to route length. This suggests that the cost of site investigation is low in Singapore, although another factor may be the limited amount of laboratory and field testing carried out in the boreholes. Average costs per meter of borehole, including mobilisation and all associated testing was as low as $138/m on some sections of the work, or just over $4,000 for a 30m deep borehole. The low cost structure of the Singapore investigation industry is largely due to the use of imported contract labour, brought in from countries such as India, Bangladesh and Burma, to work the rigs. The labour typically has no prior experience of the construction industry, and the workers are mainly trained on the job in Singapore. These workers are generally allowed to serve only two, three year, contracts in Singapore. The low cost of site investigation therefore comes with a penalty in that there is a constant turnover in the drillers and their assistants. The quality of the resulting investigation is generally lower than could be obtained with a more stable workforce, unless a high level of supervision is provided.

The ratio of claims to civil cost was four times higher for the bored tunnelling compared with the cut-and-cover excavations. It was therefore decided to focus the detailed review on the claims resulting from bored tunnelling.

As shown in Table 3, the average spacing of boreholes, at time of tender, along the bored tunnels was 39m. In all cases, the contractors carried out a significant programme of further investigation after the award of contract, in order to justify and refine their designs and selection of construction methods. Final borehole spacing along the bored tunnels averaged less than 30m. A common comment from outsiders has been the generous provision of boreholes, in comparison with other major projects both in Singapore and outside.

The claims lodged were separated into two groups: those resulting from different strata being encountered, and those that resulted from the behaviour of the strata being different to that anticipated. Table 4 separates the claims on this basis.

Table 4. Basis for claims, bored tunnels

	Amount claimed S$ Million	Percentage %
Strata incorrectly identified	4.6	12
Strata correctly identified, but behaviour different	33.8	88
Total	38.4	100

It can be seen that nearly ninety percent of the claims, by value, were due to the behaviour of the ground being different to what was anticipated, even though the geological stratum had been correctly identified from the boreholes. This would suggest that the number and spacing of boreholes was generally adequate, but that there was a deficiency in some aspects of the testing that was carried out to assess the likely response of the ground to tunnelling.

Twelve percent of the claims, by value, related to encountering strata different to that identified in the boreholes. These claims all related to the Fort Canning Boulder Bed, which was found in just 660m of

the tunnelling. This deposit was not identified in the geological map of Singapore. On the NEL the boulder bed was found in areas outside those where it had previously been mapped by Shirlaw et al (1990) and Han et al (1994). In particular, the Boulder Bed was found mantling the south-east slope of Fort Canning Hill, and at the eastern foot of Mount Emily, under and near the Bukit Timah canal. The presence of the Boulder Bed on the face of Fort Canning Hill is consistent with the unit being of colluvial origin.

4.2 Detailed Assessment

The tunnelling for the NEL was carried out under 6 major civil contracts. Each tunnelling contractor undertook between two and four tunnel drives. There were a total of 11 sections of twin, parallel tunnel constructed. The site investigation and the claims for unforeseen ground conditions were tabulated in relation to contract number, drive and ground conditions encountered. The data so assembled was used to identify whether there was a correlation between a number of factors and the value of the claims made.

The factors investigated for correlation with claims were;

- The amount spent on site investigation.
- The length of borehole drilled
- The proportion of boreholes terminated above tunnel level
- The geological strata encountered in the tunnel
- The tunnelling method
- The contractor's tender price
- The 'claims consciousness' of the contractor

4.2.1 The amount spent on site investigation.

Although the amount spent on site investigation was initially reviewed for each construction contract, a more detailed assessment was made by reviewing the investigation cost and claims for each pair of tunnel drives. Figure 5 presents the claims plotted against the cost of the site investigation for each section of tunnelling, with both figure normalised by the cost of the tunnelling. It can be seen that almost all of the claims occurred on three of the twelve sections of work. However, these three sections were also the three sections where the greatest proportional amount had been spent on site investigation.

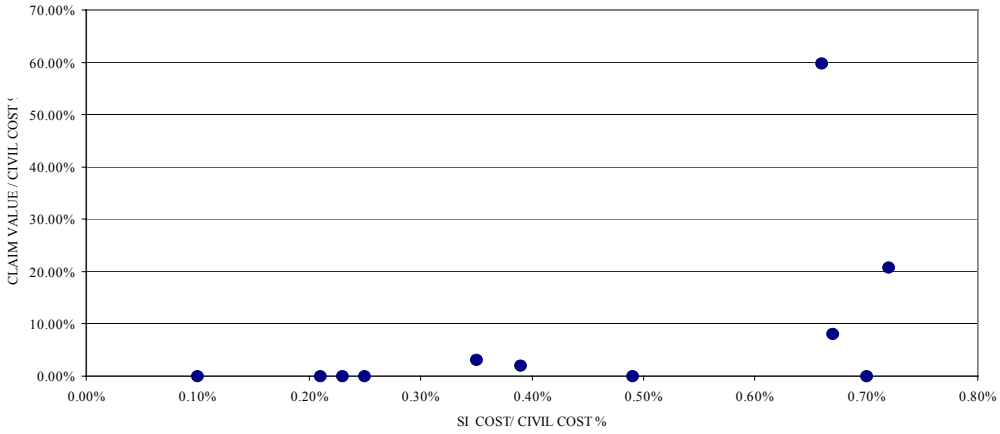


Figure 5. Relative cost of site investigation compared with relative amount claimed for each tunnelling section of the NEL.

4.2.2 The length of borehole drilled.

Figure 6 shows the claims (as a proportion of civil cost) plotted against the ratio of borehole length drilled to tunnel length driven for each of the six construction contracts involving bored tunnelling.

There is no clear relationship within the NEL data between these ratios. Also plotted on figure 6 is the suggested average relationship between these ratios as proposed by Westland et al. (1998).

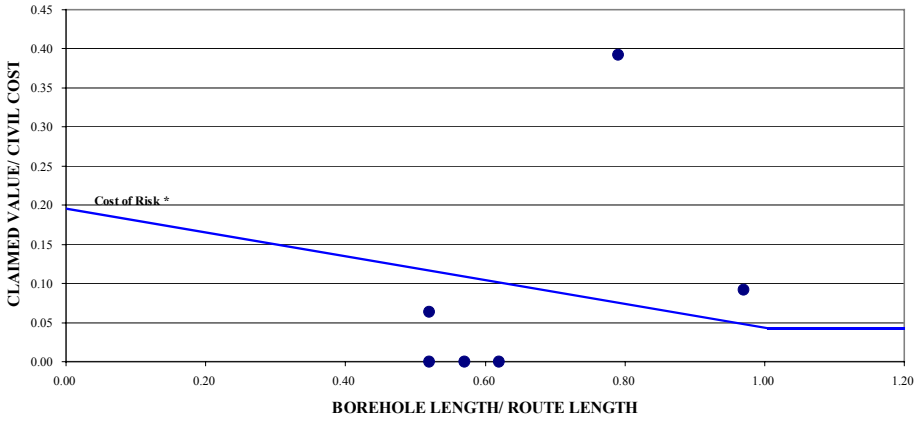


Figure 6. Relative value of claims compared with the ratio of borehole length to route length for each tunnelling contract

4.2.3 The proportion of boreholes terminated above tunnel level

Figure 7 presents the relationship between the value of the claims (as a proportion of the construction cost) and the proportion of boreholes that were taken to tunnel crown level or deeper. It can be seen that on three sections of the work a significant proportion of boreholes were terminated above the level of the tunnel. This only occurred where the rock grades of weathered Jurong formation or Bukit Timah granite were encountered. It can be seen that there is a strong correlation between the value of the contractor's claims and the proportion of the boreholes that were not taken to tunnel level.

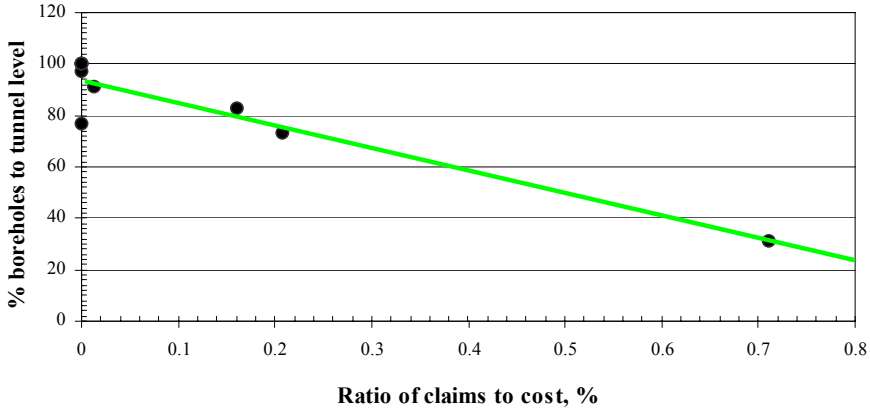


Figure 7. Relative value of claims compared with proportion of boreholes reaching tunnel level.

4.2.4 Geological unit encountered

The NEL tunnels encountered five major geological units:

- The soft soils of the Kallang formation (Holocene and late Pleistocene age)
- The cemented soils/weak rocks of the Old Alluvium, OA (Pleistocene age)
- The colluvium of the Fort Canning Boulder Bed, FCBB (Pleistocene age)
- The sedimentary rocks of the Jurong Formation (Upper Triassic to Early Jurassic)
- The igneous rocks of the Bukit Timah granite (Lower to Middle Triassic)

The total length of tunnel on the NEL was divided into six broad categories: the five major geological units listed above and a category to cover a mixed face of Kallang Formation soils and the stronger underlying units. The claims were then apportioned between these six categories. The value of the claims per kilometer of tunnel driven was then calculated for each of the categories. This is presented

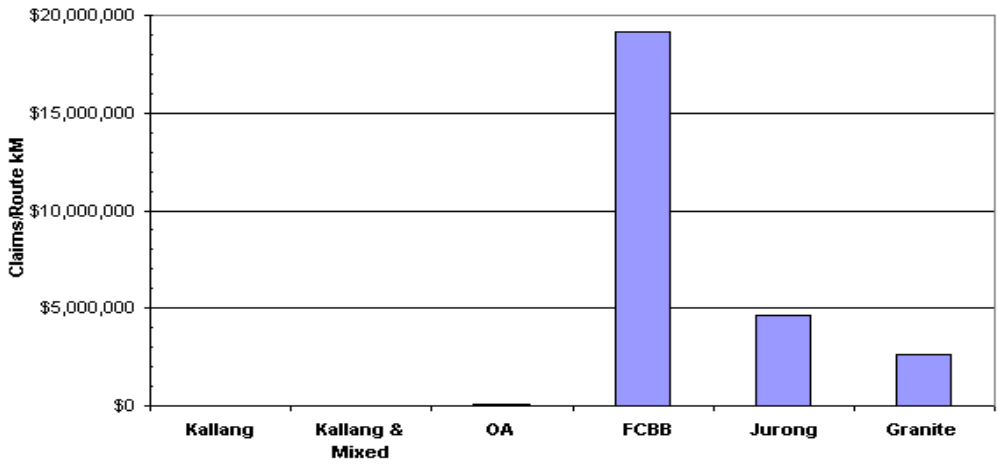


Figure 8. Value of claims per route kilometer in the main strata encountered

in Figure 8. The bar chart shows that the claims were predominantly in the rock strata, rather than in the softer ground.

From figure 8 it is apparent that the claims were disproportionately concentrated in the rock strata, and in the Fort Canning Boulder Bed. Most of the claims in relation to the Fort Canning Boulder Bed were in relation to the strength of the quartzite boulders found in the bed. It would appear that the amount of information obtained on these strata, and in particular the nature of the rocks and weathered rocks, may not have been adequate to fully characterise the conditions to be encountered. To assess this, the cost of the laboratory testing undertaken for soil and rock was calculated, and compared with the proportion of the tunnelling in soil and rock, based on whether the material in the face would be tested using soil or rock testing. This is shown in Table 5.

Table 5. Cost of laboratory testing, separated between soil and rock testing

	Soil	Rock	Total	Rock, percentage of total
Testing Cost, $	215,445	6,598	222,043	3.1
Bored tunnel, m	13,100	9,700	22,800	42.5

It can be seen that the laboratory testing was concentrated disproportionately on the soil, rather than the rock, making it understandable why the claims show the opposite pattern, with the claims being concentrated disproportionately in the rock strata.

4.2.5 Tunnelling Method

Both EPB and open face shields were used for the construction of the NEL. The open-face shields were used in only two geological units: the Fort Canning Boulder Bed and the Jurong Formation. The tunnelling using EPB shields also encountered these units, so a direct comparison can be made between the two tunnelling methods in similar ground conditions. Figure 9 shows the results of this analysis.

For the Fort Canning Boulder Bed, there were relatively high claims in relation to the open-face shields, but no claims associated with EPB tunnelling. For the Jurong Formation, the claims have been further subdivided in claims that the material was too unstable (i.e. too weak) and claims that the material was too difficult to excavate (i.e. too strong). It can be seen that there was a big difference in the claims attributed to the ground being weaker and less stable than anticipated. The open-face shield

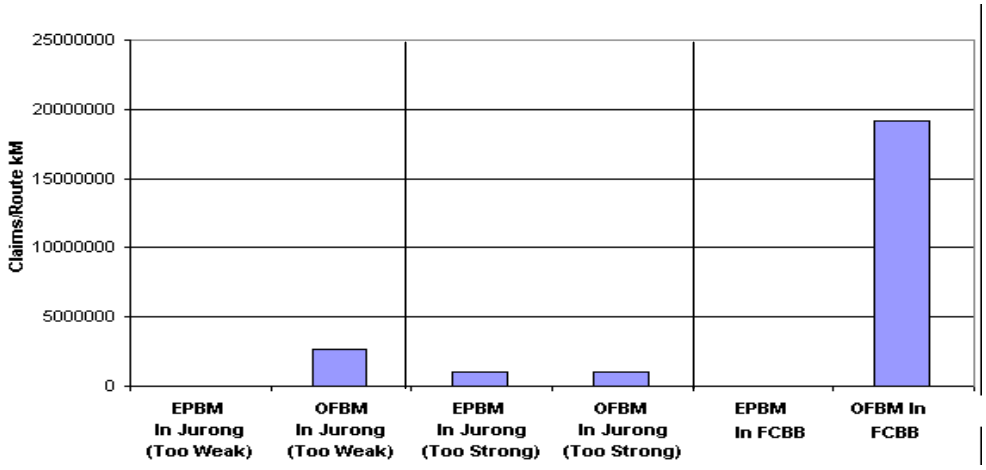


Figure 9. Comparison between claims per route kilometer in strata where both Earth pressure Balance (EPBM) and open-face machines (OFBM) were used.

tunnelling was adversely affected in these circumstances, as the poor face stability resulted in slow progress and the need to take additional measures, such as grouting, to improve stability. In contrast, similar conditions on the EPB drives do not appear to have had a significant impact on the tunnelling, and no claims were made on this basis. There was an almost identical value of claims relating to the ground being stronger than anticipated, per kilometer of tunnel, for both the EPB and open-face shield tunnelling. A mixed face of strong rock and very weak, highly or completely weathered, rock posed a significant problem for EPB tunnelling (Shirlaw et al. 2000). Attempting to maintain a high face pressure to maintain stability in the weaker, less stable materials resulted in high rates of abrasion to the cutting tools and other parts of the shield exposed to the ground and spoil. Maintainance of the pressures, and of face stability, was not always possible (Shirlaw 2002). In Figure 10, the ratio of claims to contract value is presented for the EPB driven tunnels, separated by geological unit. It can be seen that, broadly, the claims increased with increasing age and strength of the strata encountered. For the open face shields, the problem with strong rock related only to the time involved in breaking and removing the rock.

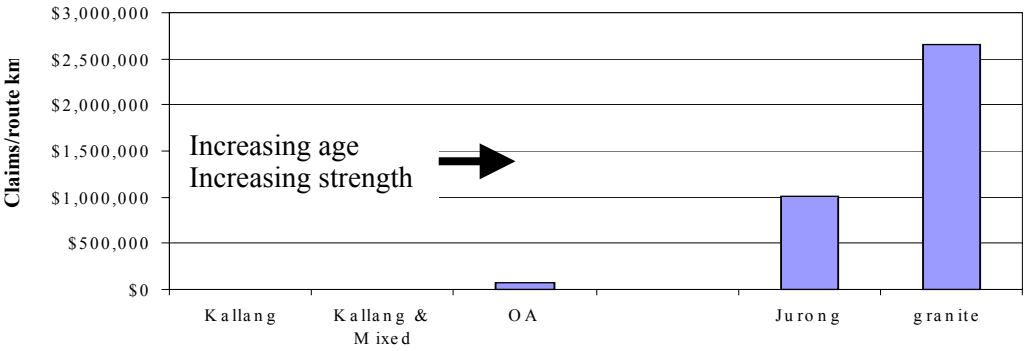


Figure 10. Value of claims per route kilometer for the tunnels driven using EPB shields, related to the different strata encountered.

4.2.6 The contractor's tender price

It is a commonly expressed view that some contractors deliberately provide low tender prices and then seek to make up the difference by means of claims. This issue does not strictly concern the planning of site investigations. However, it is worth assessing whether there was any evidence of a link between low tender prices and claims. If there was a link, then this would tend to skew the data, and this would need to be taken into account in evaluating it.

Figure 11 shows the tender prices, expressed as cost per meter of tunnel, plotted against claims made. It can be seen that there is no evidence of any link between low tender prices and the value of the

claims.

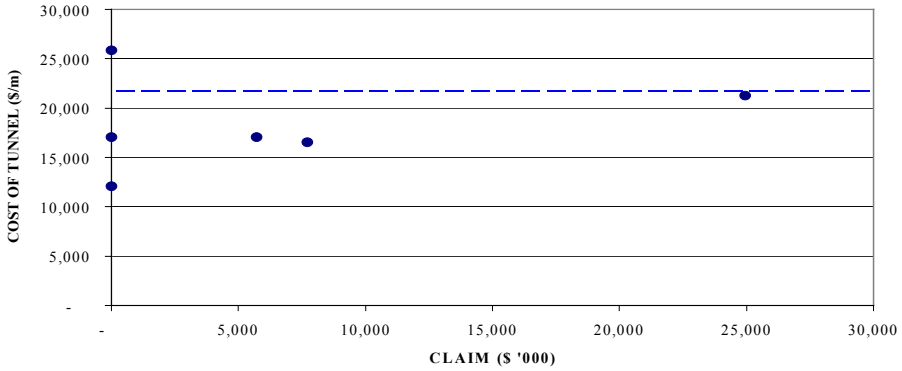


Figure 11. Comparison of the tender unit cost of building the tunnels plotted against the total amount of the claims made by each contractor

4.2.7 The 'Claims consciousness' of the contractor

Just as low tender prices are often asserted to be a factor in claims, it is also commonly asserted that some contractors are more 'claims conscious' than others. This could, if true, skew the data. The 'claims consciousness' of the contractors was assessed by comparing the ratio of claims to civil cost for the tunnelled sections with the ratio for the cut-and-cover sections. Six of the civil construction contracts included both bored tunnelling and cut-and-cover construction. For these six contracts a direct comparison could be made of the relative amounts claimed for the different methods of construction.

If any of the contractors were highly 'claims conscious', then it would be likely that there would be a similar distribution of claims for both methods of construction. As can be seen in Figure 12, one of the six contractors might have been 'claims conscious', but there is no evidence that the other five were.

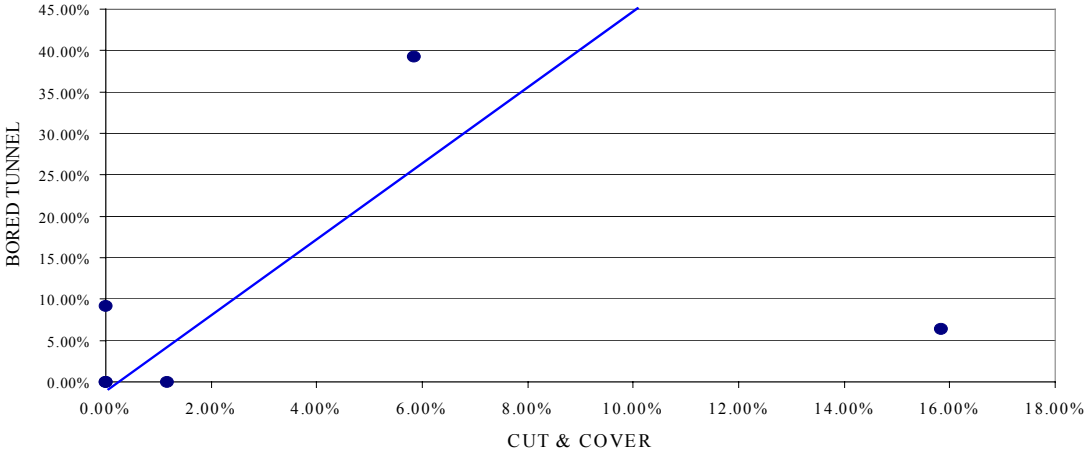


Figure 12. The ratio of the total amount claimed to the value of the civil works, for both bored tunnelling and cut and cover works, for each contract where both types of work were carried out.

5 DISCUSSION

Based on figures 11 and 12, there is little evidence that the data on claims was artificially inflated due to deliberate tender strategy or particular 'claims consciousness'. This paper cannot comment on, or assess, the validity or true value of the individual claims lodged. However, the general trends identified in the data can be used to assess where additional effort could be usefully directed in future site investigation programmes for tunnelling in Singapore.

As discussed above, the overall number of boreholes drilled for the NEL appears to have been adequate to correctly identify the stratigraphy along the route, apart from very localised areas where the Fort Canning Boulder Bed was present. The majority of the tunnelling for the line was carried out using EPB shields. From the data, the major problems were encountered in the weathered rocks of the Jurong Formation and the Bukit Timah granite. There were no claims in relation to this method when tunnelling through the soft soils of the Kallang Formation, and only minor claims in relation to tunnelling in the Old Alluvium. Improving the quantity and quality of the site investigation in the weathered rocks is therefore suggested.

The clearest correlation between any of the factors studied was between the value of claims and the proportion of boreholes taken to tunnel level. In certain areas, 20% to 60% of the boreholes drilled were terminated above the tunnel. This particularly occurred where the tunnels were under hills, where even boreholes longer than the average might not reach tunnel level. These boreholes were typically terminated when rock was first encountered. As a result of the termination, no information was obtained at that location on the nature of the rock that the tunnel would actually be driven through. This may help to explain the lack of correlation between claims and both costs and borehole length shown in figures 5 and 6. The highest costs and longest boreholes were in the areas under hills, but many of the boreholes were not taken to the crucial level: that of the tunnel.

Reviewing the claims that were lodged in respect of the EPB tunnelling in weathered rock, these principally involved:

- The stability of mixed grades of weathered rock
- Slow rates of advance in the strong and very strong rock
- Abrasion/damage to the shield when excavating in strong or very strong rock and in mixed soil and rock grades of weathered rock

Improving the site investigation to allow more accurate characterisation of the weathered rock strata along the alignment should help to reduce the risk of encountering unforeseen conditions. It is suggested that the following improvements can be made:

- More, and more careful, coring of the weathered rock
- Ensuring careful description and classification of the weathered rock recovered
- Increasing the number of permeability tests carried out
- Increasing the number of tests carried out for the strength and abrasivity of the rock
- Use of geophysical methods to provide information on structural features within the rock.

At the start of this paper, reference was made to recommendations made by the U.S. National Committee on Tunnelling Technology. The committee recommended that 1.5m of borehole is drilled for every metre of tunnel, and that site investigation should be about 3% of project cost. The respective amounts for the NEL tunnels were 0.6m/m, and 0.422%. Deepening the boreholes and increasing the amount of rock testing, as suggested here, would have increased both values, but would still have left both significantly below the U.S. committee recommendations. Based on Figure 6, there is no strong relationship between the borehole length and the amount of claims. Ensuring that the maximum amount of relevant information is obtained from those boreholes that are drilled appears to be as least as important as the absolute number of boreholes.

In the assessment, no account has been taken of existing borehole data, and of the experience gained of the behaviour of the various soil and rock strata during previous underground construction in Singapore. Clearly, this knowledge, where available, forms a valuable supplement to the borehole programme carried out for the line. This would help to explain why the investigation programme for the

NEL appears to have been generally adequate. Westland et al. refer to borehole coverage for the Sheppard Line in Toronto with a borehole length of 0.8m/metre of twin tunnel, which also appears to have been adequate for tendering purposes. It seems that for shallow tunnels in urban areas with a previous, published, history of underground construction a ratio of between 0.5 and 0.8m/metre of twin tunnel is normally adequate.

As explained above, the cost structure of the Singapore site investigation industry is low, and this may explain much of the discrepancy between the U.S. recommendations and the achieved figure for the NEL. Allowing for some additional costs in relation to additional borehole length and rock testing, it would appear that a cost of site investigation of between 0.5% and 1% of the cost of constructing the tunnels should generally be adequate, in Singapore, to control the risk of encountering unforeseen ground conditions.

One particular factor that this study has highlighted is the difference between open face shields and EPB shields in terms of the sensitivity of the method to the type of ground conditions. With the open face shields, problems (and claims) can develop in soft ground or rock. Problems with the earth pressure balance shields were largely restricted to rock or mixed faces of soil and rock. Overall, it appears that there is a higher risk of claims with open face shields than earth pressure balance shields.

6 CONCLUSIONS

Based on data from the NEL it appears that a new, pre-tender, borehole programme of between 0.5m to 0.8m/m of twin tunnel, and a site investigation cost of 0.5% to 1% of tunnelling cost is generally adequate to keep claims to a low level. This is a lower degree of site investigation than recommended by the U.S. National Committee on Tunnelling Technology. This can be explained by the fact that, in Singapore, there is a considerable database of existing borehole records and of published information on ground behaviour during underground construction, which augment the information from any new investigation programme.

The data shows that obtaining the maximum amount of information from those boreholes that are drilled is at least as important as the absolute meterage of borehole drilled. It is essential that all boreholes go to, and below, the tunnel level. For tunnelling using EPB machines in varied conditions, the investigation must provide sufficient information on the presence and nature of both soil and rock along the alignment, and on the location and nature of mixed rock/soil interfaces.

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